Analog Circuits Course Project

Course: Analog Circuits

Course Code: BECE206L

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Semester: Fall Sem – 2025-26

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Abstract

This report details the design, implementation, and testing of a low-cost, ESP32-based IoT monitoring system for standard refrigerators. The system is designed as a non-invasive, retrofit device that actively monitors environmental parameters to prevent energy waste, ensure food safety, and provide critical gas leak alerts.

The prototype integrates three distinct sensors: a Light Dependent Resistor (LDR) to detect the door's open/closed status, an MQ-3 alcohol sensor for food spoilage detection (fermentation), and an MQ-2 sensor for household gas leak detection. Sensor data is processed locally by the ESP32, which is programmed to identify and prioritize three distinct alarm states.

Data is continuously logged to the ThingSpeak cloud platform for real-time data visualization. Upon detecting a critical event, the system uses ThingSpeak's "React" and "ThingHTTP" applications to send a specific, prioritized push notification directly to a user's mobile device via the Pushbullet API. This document covers the system architecture, hardware implementation, calibration, cloud integration, and demonstration results, as well as the limitations of the prototype and paths for future development.

Problem Statement

Traditional refrigerators operate as "black boxes," offering no insight into their internal environment or the status of their contents. This lack of information leads to several common and costly problems that this project seeks to address:

- **Food Waste and Safety:** Food spoilage often goes unnoticed until it is severe. This results in significant food waste and poses a health risk if spoiled items are accidentally consumed. There is no system to provide an early warning of fermentation or decay.
- **Energy Inefficiency:** A refrigerator is a major source of household energy consumption. When the door is left open accidentally, cold air escapes, forcing the compressor to run excessively to maintain temperature. This wastes a substantial amount of electricity.
- **Lack of Remote Information:** Users have no way of knowing if their appliance is functioning correctly or if a problem (like a door left ajar) has occurred when they are not home.
- **Household Safety Gaps:** The kitchen is the most common location for a household gas leak, typically from a stove. A standard refrigerator provides no sensing for this dangerous, external environmental hazard.

This project aims to solve these problems by creating a single, integrated "smart" device that provides timely alerts for energy waste, food spoilage, and gas leaks.

Project Objective

The primary technical objective is to design and build a connected, real-time monitoring system as a retrofit add-on for traditional refrigerators. This system is capable of:

- **Sensing** critical environmental parameters: door status (via internal light), spoilage-related gases (ethanol), and external gas leaks (LPG, methane).
- **Processing** sensor data locally on the microcontroller to determine if a specific alarm state is met, using pre-calibrated thresholds.
- **Actuating** both a local, audible/visual alarm (buzzer and LEDs) and a remote, instant mobile notification.
- **Logging** all sensor data to a cloud platform for live visualization and historical analysis, proving the IoT concept.

System Architecture

The project follows a standard, event-driven IoT architecture. The flow of data is linear and can be broken down into four distinct stages:

1. **Input (Sensing):** The LDR, MQ-3, and MQ-2 sensors continuously read the ambient environment. Their analog resistance changes are converted to analog voltage signals.
2. **Processing (Local):** The ESP32's Analog-to-Digital Converter (ADC) reads the voltage from all three sensors. This raw data is compared against pre-set software thresholds (e.g., SPOILAGE_THRESHOLD). The ESP32's logic also manages a timer for the door-open event and prioritizes which alarm is most critical.
3. **Output (Local Actuation):** If an alarm is triggered, the ESP32 sends a HIGH signal to the corresponding digital pins, activating the Red LED and the active buzzer. If the system is normal, the Green LED remains on.
4. **Cloud & Remote Alerting:** The ESP32 uses its built-in Wi-Fi to send all sensor data to a ThingSpeak channel. ThingSpeak logs the data and (if an alarm is detected) triggers a webhook to the Pushbullet API, which sends the final notification to the user's phone.

Hardware Components & Implementation

Components Used

- **Microcontroller:** ESP32-WROOM-32 (Chosen for built-in Wi-Fi, multiple ADC pins, and processing power).
- **Prototyping:** Solderless Breadboard (for initial circuit assembly).
- **Sensors (Input):**
  - MQ-3 Alcohol Sensor (Detects ethanol vapours for spoilage).
  - MQ-2 Gas Sensor (Detects LPG, methane, and smoke for gas leaks).
  - Light Dependent Resistor (LDR) (Detects internal fridge light for door status).
- **Actuators & Indicators (Output):**
  - Active Buzzer (Provides an audible alarm).
  - 5mm Red LED (Alarm status) & Green LED (System OK status).
- **Passive Components:**
  - 1x 10kΩ Resistor (Used as a pull-down for the LDR voltage divider).
  - 2x 220Ω Resistors (Used as current-limiting resistors for the LEDs).

Circuit Design & Pin Connections

The circuit is powered by the ESP32's 3.3V and GND pins, which are connected to the breadboard's power rails. All sensors and actuators are connected as follows:

Component	ESP32 Pin	Purpose
LDR (via 10kΩ Resistor)	GPIO 34	Analog Input (Door Sensor)
MQ-3 Sensor (A0)	GPIO 35	Analog Input (Spoilage Sensor)
MQ-2 Sensor (A0)	GPIO 32	Analog Input (Gas Leak Sensor)
Green LED (via 220Ω)	GPIO 4	Digital Output (System OK)
Red LED (via 220Ω)	GPIO 2	Digital Output (Alarm Active)
Buzzer (Signal)	GPIO 27	Digital Output (Audible Alarm)

System Logic & Cloud Integration

Local Logic & Thresholds

The system logic is built around three distinct, prioritized alarms. The alarm thresholds were determined through calibration and testing.

Alarm Type	Trigger Sensor	Threshold (Variable)	Alarm Code
Gas Leak	MQ-2	>1000	3
Spoilage	MQ-3	>2000	2
Door Open	LDR	>1000 & Time > 30 sec	1

The code logic prioritizes alerts in order of severity (3 > 2 > 1) to ensure the most critical notification is sent.

Cloud Platform: ThingSpeak

ThingSpeak serves as the project's cloud backend for data logging and visualization.

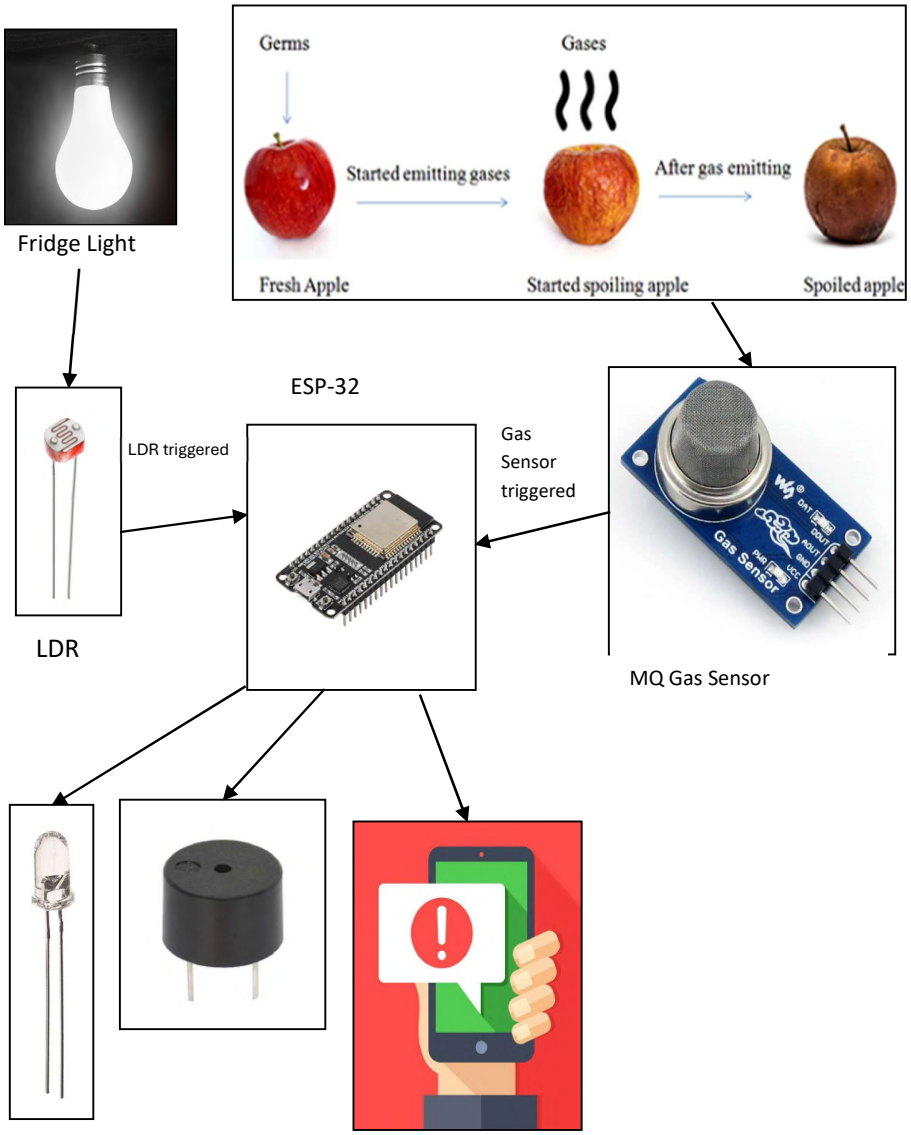
- **Mechanism:** The ESP32 sends an HTTP POST request to the ThingSpeak API every 60 seconds (or immediately if an alarm is triggered).
- **Channel Setup:** A single channel ("Smart Fridge Monitor") was created with four fields to accept the data:
    - **Field 1:** LDR Value
    - **Field 2:** Spoilage (MQ-3) Value
    - **Field 3:** Alarm Status (Sends the code 0, 1, 2, or 3)
    - **Field 4:** Gas (MQ-2) Value
- **Visualization:** The "Private View" of the channel automatically generates live-updating graphs for each field, providing a real-time dashboard of the refrigerator's environment.

Alerting System: Pushbullet

Pushbullet is used to send instant, private notifications to a mobile device.

- **Mechanism:** The **ThingSpeak "React" App** is used to monitor the "Alarm Status" field.
- **Configuration:** Three separate Reacts were created, one for each alarm code:
    1. **React 1 (Door):** Triggers if Field 3 is equal to 1.
    2. **React 2 (Spoilage):** Triggers if Field 3 is equal to 2.
    3. **React 3 (Gas Leak):** Triggers if Field 3 is equal to 3.
- **Action:** Each React is linked to a unique **"ThingHTTP"** request. This request sends an API call to Pushbullet with a specific message body (e.g., {"type": "note", "title": "🔥 GAS LEAK DETECTED!"...}). The Pushbullet app on the phone receives this call and displays the specific notification.

Visual Flowchart



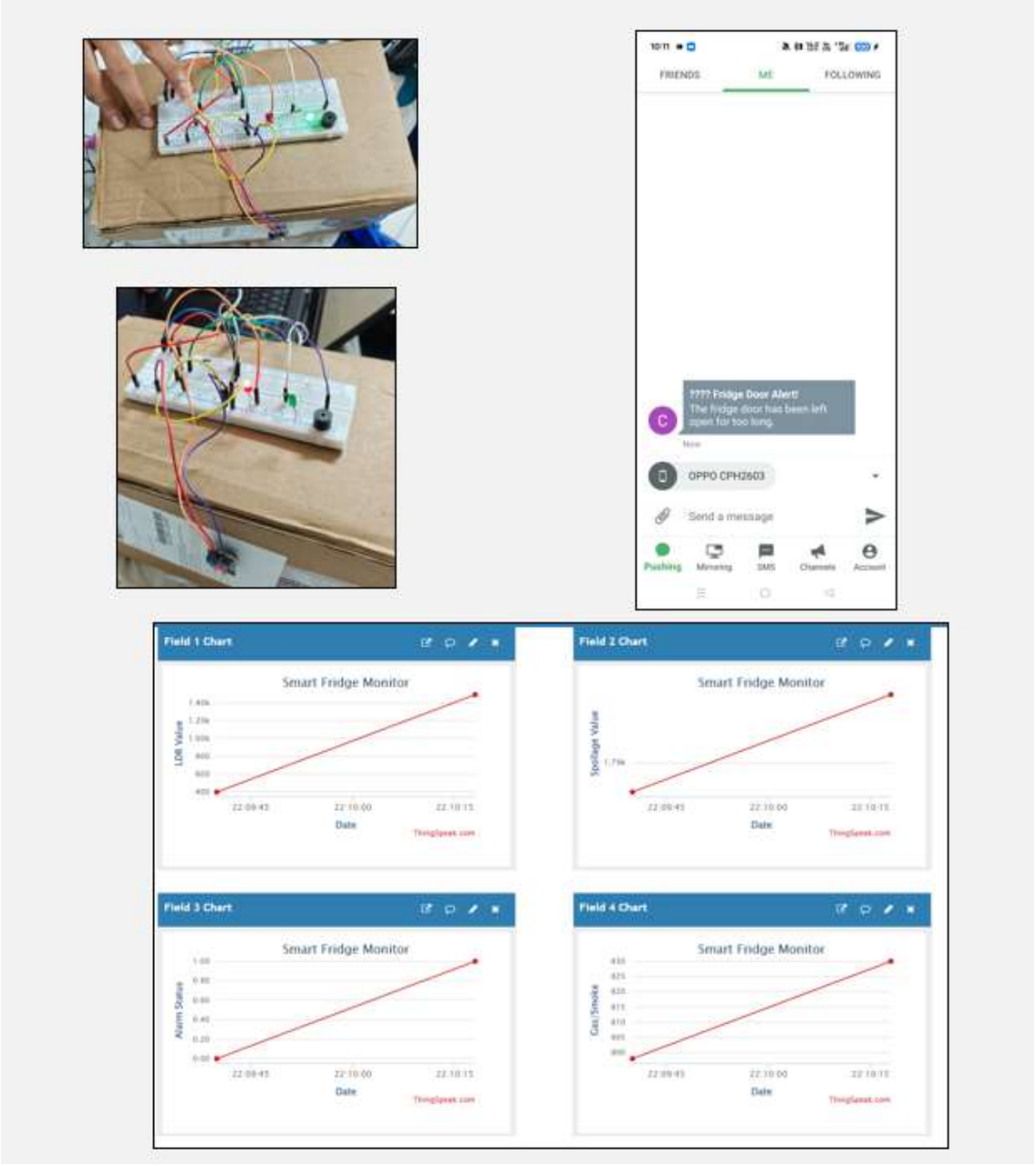
Code

https://drive.google.com/drive/folders/153u9xrw-GLW8u065T2N7MVmIp23at5tX?usp=sharing

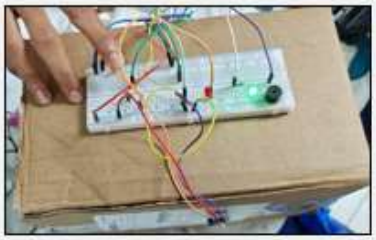
Results & Demonstration

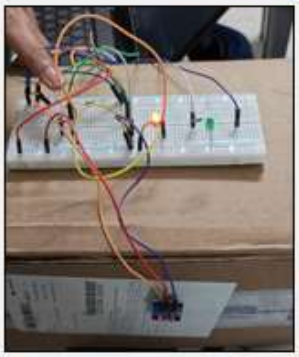
The prototype was successfully tested under three distinct alarm conditions.

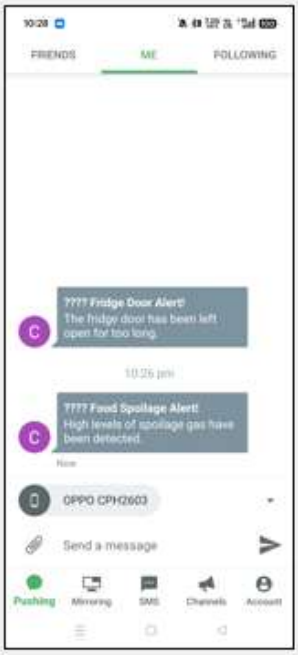
- **Test Case 1: Door Open Alarm (Alarm 1)**
  - **Action:** The LDR sensor was uncovered for over 30 seconds.
  - **Result:** The local red LED and buzzer activated. ThingSpeak Field 3 logged a 1, and a "Fridge Door Alert!" notification was received.

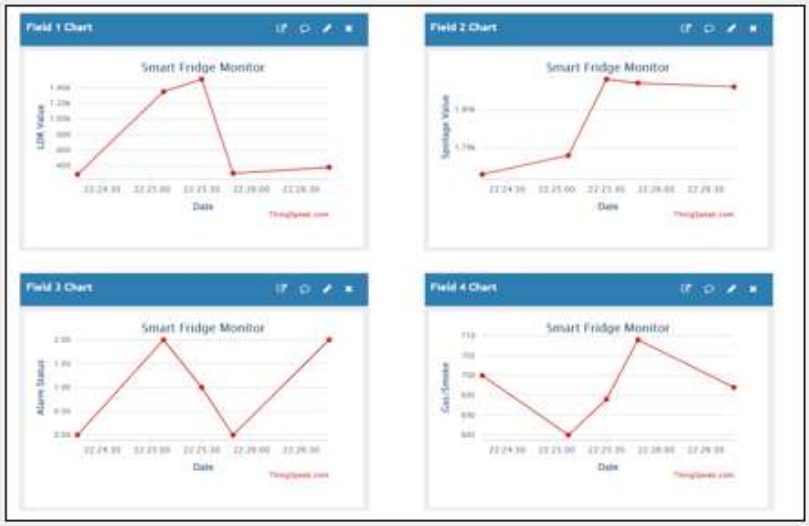


- **Test Case 2: Spoilage Alarm Test (Alarm 2)**
  - **Action:** 70% isopropyl alcohol (simulating fermentation vapor) was brought near the MQ-3 sensor.
  - **Result:** The local red LED and buzzer activated instantly. ThingSpeak Field 3 logged a 2, and a "🍎 Food Spoilage Alert!" notification was received.

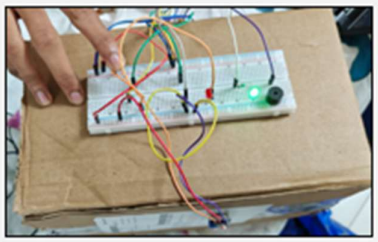


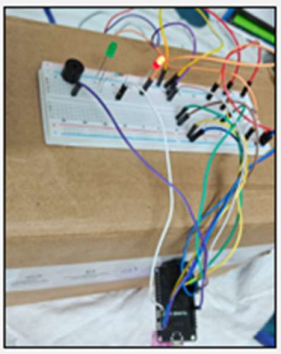


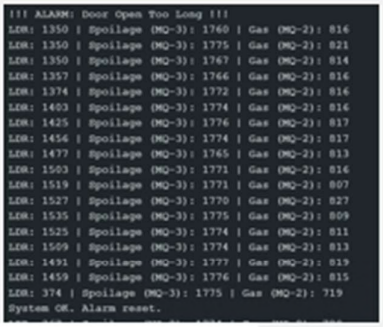


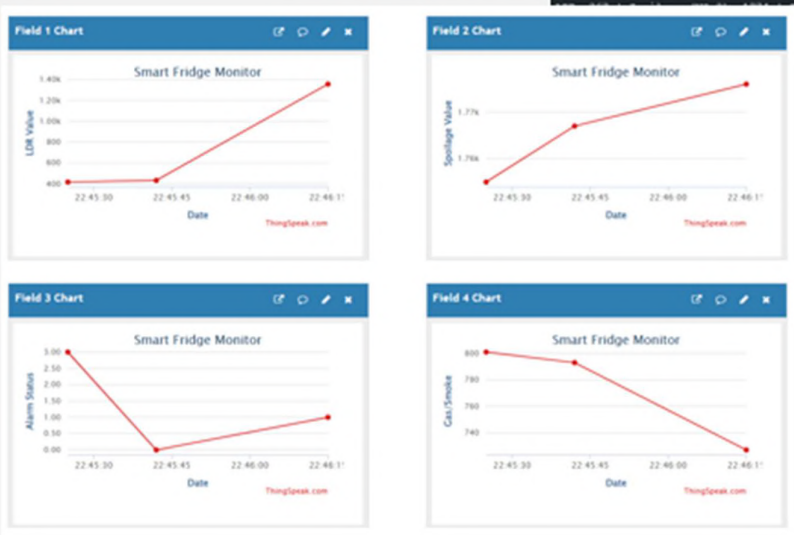


- **Test Case 3: System Reset**
    - **Action:** All alarm conditions were removed (LDR covered, vapors cleared).
    - **Result:** The local red LED and buzzer turned off. The green "System OK" LED turned on. ThingSpeak Field 3 logged a 0.









Prototype Status & Limitations

This project is a successful **functional prototype** designed to simulate the core logic of the final product. It proves the concept is viable. However, significant Research & Development (R&D) is required to address key limitations:

- **Power Supply:** A robust power solution is needed (e.g., long-life battery or hard-wiring) to replace the USB cable.
- **Physical Integration:** A final design is needed to determine how the device is mounted inside or outside the fridge for optimal sensing and durability.
- **Miniaturization:** The breadboard circuit must be redesigned as a compact, custom Printed Circuit Board (PCB).
- **Environmental Durability:** Long-term testing is required to validate sensor performance in a cold, high-humidity environment.

Conclusion

This project successfully demonstrates a viable proof-of-concept for a multi-sensor IoT monitoring system. The core objectives were met: we successfully integrated sensors to log environmental data to the ThingSpeak cloud and built an event-driven system to send immediate, prioritized mobile alerts using Pushbullet. This prototype confirms that the chosen architecture (ESP32, cloud services) is an effective and low-cost foundation for a smart monitoring solution. The project successfully simulates the system's core logic and lays the groundwork for the future R&D needed to move from a prototype to a finished product.

Future Work

The following paths are identified for moving from the current prototype to a market-ready product:

- **Sensor Enhancement:**
    1. Explore more reliable door sensors, such as a **magnetic reed switch** or **Infrared (IR) proximity sensor**, which would be unaffected by ambient light.
    2. Integrate an **MQ-135 Air Quality Sensor** to detect a wider range of spoilage-related gases, such as ammonia (from meat) and sulphides.
- **Hardware Optimization:**

    1. Investigate **lower-power Wi-Fi microcontrollers** (like the ESP8266 or ESP32-C3) to reduce energy consumption and make a long-life battery-powered version feasible.
    2. Design and 3D print a **custom, compact, and food-safe enclosure** to house the final PCB and sensors.
- **Next-Generation Sensing:**
    - Research **tailor-made sensors** (e.g., using nanotechnology) that are highly sensitive to the precise volatile organic compounds (VOCs) released by specific types of food spoilage.
- **Software & App Development:**
    - Replace Pushbullet with a **dedicated mobile application** (using a backend like Firebase) to create a professional user interface and add more features (e.g., data history, settings, and alert management).